

## **Hadoop HDFS Practice Commands**

### **Part 1: Start Hadoop Services in Ubuntu**

#### **1. Open Terminal in Ubuntu**

Press:

Ctrl + Alt + T

#### **2. Go to Hadoop sbin Folder**

```
cd /home/hadoop/hadoop/sbin
```

#### **3. Start HDFS**

```
./start-dfs.sh
```

#### **4. Start YARN**

```
./start-yarn.sh
```

#### **5. Check Running Hadoop Daemons**

```
jps
```

### **Part 2: Practice HDFS Commands**

#### **1. Create Directory (mkdir)**

Create directory in HDFS:

```
hdfs dfs -mkdir /abc
```

Check directory:

```
hadoop fs -ls /
```

#### **2. Upload File Using put**

First create a text file.

Create file:

```
gedit sample.txt
```

Paste the following text into the file:

```
hi how are you  
how is your job  
how is your family  
how is your brother  
how is your sister
```

Save the file.

Upload file to HDFS:

```
hdfs dfs -put sample.txt /abc/WC.txt
```

Check:

```
hdfs dfs -ls /abc
```

### **3. Display File (cat)**

```
hdfs dfs -cat /abc/WC.txt
```

### **4. copyFromLocal**

```
hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt
```

Check file content:

```
hdfs dfs -cat /abc/WC2.txt
```

### **5. get (Download from HDFS)**

```
hdfs dfs -get /abc/WC.txt ~/Downloads/WWC.txt
```

### **6. getmerge**

Merge two HDFS files into one local file:

```
hdfs dfs -getmerge /abc/WC.txt /abc/WC2.txt ~/Desktop/Merge.txt
```

### **7. copyToLocal**

hdfs dfs -copyToLocal /abc/WC.txt ~/Desktop

## 8. Move Directory

hadoop fs -mv /abc /FFF

Check:

hadoop fs -ls /FFF

## 9. Copy Directory

hadoop fs -cp /FFF /LLL

```
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~$ cd /home/hadoop/hadoop/sbin
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ ./start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 13038. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 13228. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmsce-HP-Ellite-Tower-600-G9-Desktop-PC]
bmsce-HP-Ellite-Tower-600-G9-Desktop-PC: secondarynamenode is running as process 13538. Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ ./start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 13877. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 14043. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ jps
13538 SecondaryNameNode
15538 Jps
13877 ResourceManager
14043 NodeManager
13228 DataNode
13038 NameNode
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -mkdir /abc
mkdir: /abc: File exists
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls
ls: '.': No such file or directory
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /
Found 11 items
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:43 /FFF
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:56 /LLL
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:54 /abc
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:55 /hadoop_lab
drwxr-xr-x - hadoop supergroup 0 2026-02-25 16:05 /inputbda
drwxr-xr-x - hadoop supergroup 0 2026-02-25 16:06 /outputbda
drwxr-xr-x - hadoop supergroup 0 2024-09-10 12:56 /pqr
drwxr-xr-x - hadoop supergroup 0 2025-05-07 12:26 /rgs
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:50 /test
drwxr-xr-x - hadoop supergroup 0 2025-05-27 15:31 /user
drwxr-xr-x - hadoop supergroup 0 2024-09-10 13:19 /xyz
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ gedit sample.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -put sample.txt /abc/WC.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -ls /abc
Found 3 items
-rw-r--r-- 1 hadoop supergroup 89 2026-03-06 15:41 /abc/WC.txt
-rw-r--r-- 1 hadoop supergroup 17 2026-02-25 15:53 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup 17 2026-02-25 15:54 /abc/file_cp_local.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC.txt
ht how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC2.txt
ht how are you
how is your job
how is your family
```

```
14043 NodeManager
13228 DataNode
13038 NameNode
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -mkdir /abc
mkdir: /abc: File exists
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls
ls: '.': No such file or directory
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /
Found 11 items
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:43 /FFF
drwxr-xr-x - hadoop supergroup 0 2025-04-16 11:56 /LLL
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:54 /abc
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:55 /hadoop_lab
drwxr-xr-x - hadoop supergroup 0 2026-02-25 16:05 /inputbda
drwxr-xr-x - hadoop supergroup 0 2026-02-25 16:06 /outputbda
drwxr-xr-x - hadoop supergroup 0 2024-09-10 12:56 /pqr
drwxr-xr-x - hadoop supergroup 0 2025-05-07 12:26 /rgs
drwxr-xr-x - hadoop supergroup 0 2026-02-25 15:50 /test
drwxr-xr-x - hadoop supergroup 0 2025-05-27 15:31 /user
drwxr-xr-x - hadoop supergroup 0 2024-09-10 13:19 /xyz
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ gedit sample.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -put sample.txt /abc/WC.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -ls /abc
Found 3 items
-rw-r--r-- 1 hadoop supergroup 89 2026-03-06 15:41 /abc/WC.txt
-rw-r--r-- 1 hadoop supergroup 17 2026-02-25 15:53 /abc/file.txt
-rw-r--r-- 1 hadoop supergroup 17 2026-02-25 15:54 /abc/file_cp_local.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC.txt
ht how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC2.txt
ht how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -get /abc/WC.txt ~/Downloads/WC.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -getmerge /abc/WC.txt /abc/WC2.txt ~/Desktop/Merge.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyToLocal /abc/WC.txt ~/Desktop
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -mv /abc /FFF
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /FFF
Found 5 items
-rw-r--r-- 1 hadoop supergroup 48 2025-04-16 11:43 /FFF/WC.txt
drwxr-xr-x - hadoop supergroup 0 2026-03-06 15:42 /FFF/abc
-rw-r--r-- 1 hadoop supergroup 48 2024-09-10 13:02 /FFF/WC.txt
-rw-r--r-- 1 hadoop supergroup 48 2024-09-10 13:02 /FFF/WC1.txt
-rw-r--r-- 1 hadoop supergroup 96 2024-09-10 13:22 /FFF/WC3.txt
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -cp /FFF /LLL
hadoop@bmsce-HP-Ellite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
```

## BDA Lab: WordCount MapReduce Program Execution

### Part 3: Create Java Classes in Eclipse

- Classes created:
  1. WCMapper.java
  2. WReducer.java
  3. WCDriver.java
- Code pasted from lab manual.

### Part 4: Create Input File

gedit sample.txt

#### Content of **sample.txt**:

hi how are you  
how is your job  
how is your family  
how is your brother  
how is your sister

Save the file.

### Part 5: Upload Input File to HDFS

```
hadoop fs -mkdir /inputbda
# Output: mkdir: `/inputbda': File exists
hadoop fs -copyFromLocal sample.txt /inputbda
# Output: copyFromLocal: `/inputbda/sample.txt': File exists
hadoop fs -ls /inputbda
```

#### HDFS directory content:

```
Found 1 items
-rw-r--r--  1 hadoop supergroup      89 2026-02-25 16:05 /inputbda/sample.txt
```

### Part 6: Run WordCount MapReduce

```
hadoop jar /home/hduser/Desktop/wordcount.jar WCDriver /inputbda/sample.txt /outputbda
# Output: JAR does not exist or is not a normal file (ignore if path corrected)
```

Check jar file:

```
ls -l ~/Desktop/wordcount.jar
```

### **Output:**

```
-rw-rw-r-- 1 hadoop hadoop 2606 Mar 6 16:05 /home/hadoop/Desktop/wordcount.jar
```

Check jar contents:

```
jar tf /home/hadoop/Desktop/wordcount.jar
```

### **Output:**

```
META-INF/MANIFEST.MF
WCMapper.class
WCDriver.class
WCDReducer.class
```

## **Part 7: View Output**

Check HDFS output directory:

```
hadoop fs -ls /outputbda
```

### **Output:**

```
Found 2 items
-rw-r--r-- 1 hadoop supergroup      0 2026-02-25 16:06 /outputbda/_SUCCESS
-rw-r--r-- 1 hadoop supergroup    69 2026-02-25 16:06 /outputbda/part-00000
```

Display WordCount results:

```
hadoop fs -cat /outputbda/part-00000
```

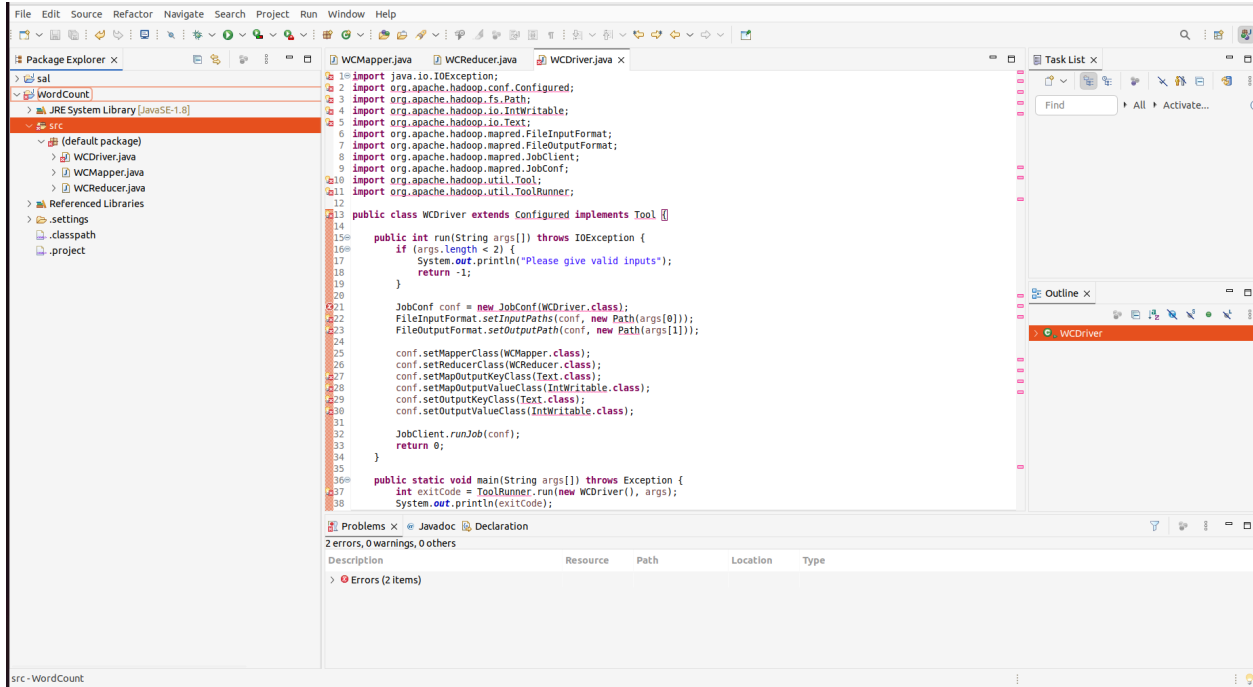
### **Output:**

```
are      1
brother  1
family   1
hi        1
how      5
```

|        |   |
|--------|---|
| is     | 4 |
| job    | 1 |
| sister | 1 |
| you    | 1 |
| your   | 4 |

Merge output to local file:

```
hadoop fs -getmerge /outputbda ~/Desktop/wordcount_result.txt
```



```
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ gedit sample.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -mkdir /inputbda
mkdir: '/inputbda': File exists
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -copyFromLocal sample.txt /inputbda
copyFromLocal: '/inputbda/sample.txt': File exists
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -ls /inputbda
Found 1 items
-rw-r--r-- 1 hadoop supergroup 89 2026-02-25 16:05 /inputbda/sample.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop jar /home/hduser/Desktop/wordcount.jar WCDriver /inputbda/sample.txt /outputbda
JAR does not exist or is not a normal file: /home/hduser/Desktop/wordcount.jar
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ ls -l ~/Desktop/wordcount.jar
-rw-rw-r-- 1 hadoop hadoop 2606 Mar  6 16:05 /home/hadoop/Desktop/wordcount.jar
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -ls /outputbda
Found 2 items
-rw-r--r-- 1 hadoop supergroup 0 2026-02-25 16:06 /outputbda/_SUCCESS
-rw-r--r-- 1 hadoop supergroup 69 2026-02-25 16:06 /outputbda/part-00000
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ jar tf /home/hadoop/Desktop/wordcount.jar
META-INF/MANIFEST.MF
WCDriver.class
WCDriver.class
WCDriver.class
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -cat /outputbda/part-00000
are
1
brother 1
fantly 1
hl 1
how 5
is 4
job 1
sister 1
you 1
your 4
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ hadoop fs -getnerge /outputbda ~/Desktop/wordcount_result.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $
```